

Addendum H — Measurement and AI-Augmented Enforcement

The Exploitation Harm Index: Technical Specification,
AI Monitoring Architecture, and Cross-Domain Enforcement Communication

Parent Document S.E.P.A. Discussion Draft v1.0	Companion Addenda Sector Guidance A–F; Individual Citizens G
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Addendum Sections

- H.1 — The Exploitation Harm Index: Full Technical Specification
- H.2 — The Four-Layer AI Monitoring Architecture
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NOTICE: This addendum is a discussion draft companion to S.E.P.A. v1.0 and carries identical advisory status.

ADDENDUM H — MEASUREMENT AND AI-AUGMENTED ENFORCEMENT

The core statute of S.E.P.A. establishes a strict-liability prohibition on exploitation and creates the Office of Harm Measurement (OHM) as the institutional body responsible for enforcing it. In its Drafting Notes, the statute identifies the Exploitation Harm Index as 'not a detail — it is the technical infrastructure upon which this entire framework rests.' The statute is explicit: 'Investment in the EHI is investment in the enforceability of the entire Act.'

This Addendum delivers on that commitment. It provides the full technical specification of the EHI, the architecture of the AI-augmented monitoring system through which the OHM will detect and measure exploitation at the scale and velocity that contemporary society produces it, the requirements engineering standards that govern both the monitoring AI and the platforms it monitors, the safeguards that prevent the monitoring architecture from becoming an instrument of the capture it was designed to resist, and the communication protocols through which violations are escalated from detection to enforcement action.

The framework developed here draws on established systems engineering practice — including Safety-Controlled Systems Analysis (STPA/STAMP), holonic architecture principles, semantic gateway design, uncertainty quantification aligned with the Guide to the Expression of Uncertainty in Measurement (GUM), and Statistical Process Control (SPC) as the operational basis for anomaly detection thresholds. These are not novel theoretical constructs. They are proven engineering methodologies applied to the specific governance challenge of exploitation detection and measurement at the societal scale.

FOUNDATIONAL DESIGN CONSTRAINT

The monitoring and measurement architecture described in this Addendum must satisfy a constraint that has no equivalent in conventional systems engineering: it must be more resistant to capture, manipulation, and abuse than the exploitation it is designed to detect. An AI monitoring system that can be gamed by the Actors it monitors, captured by the institutions that fund it, or weaponized against legitimate collective action is worse than no monitoring system at all. Every design choice in this Addendum is evaluated against this constraint first.

H.1 — THE EXPLOITATION HARM INDEX: FULL TECHNICAL SPECIFICATION

SEPA Reference	S.E.P.A. Section 2.7; Drafting Note — EHI as Load-Bearing Member
Core Function	Standardized, peer-reviewed, publicly accessible instrument for quantifying exploitation harm across all domains
Scope	All Actors; all harm categories under Section 2.2; all addendum domains A–G

H.1.1 — The Five Scoring Dimensions

The EHI is a composite score computed across five orthogonal dimensions, each independently measurable and collectively sufficient to characterize the full harm profile of an exploitative act. The five dimensions correspond to the fundamental questions that any harm assessment must answer. They are designated using the structured inquiry framework WHO / WHAT / WHEN / WHERE / WHY — not as rhetorical categories but as precision measurement axes.

Axis	Dimension	Label	What It Measures	Example Metric
WHO	Breadth	B	Number and vulnerability profile of persons, communities, or natural systems harmed	Affected population size, demographic vulnerability index, and ecosystem coverage area
WHAT	Magnitude	M	Severity of the harm per affected unit — the depth of damage to an individual, community node, or ecosystem component	Clinical harm severity scale; financial loss per capita; biodiversity loss index
WHEN	Duration	D	Temporal extent of the harm: how long has it persisted, and at what trajectory	Harm-years; onset-to-detection lag; acceleration coefficient
WHERE	Concentration	C	Geographic, demographic, and systemic distribution — whether harm is diffuse or targeted at specific Vulnerable Parties	Gini coefficient of harm distribution; spatial clustering index; demographic targeting ratio
WHY	Reversibility	R	Whether and to what degree the harm can be remediated, the permanent loss component of the total harm	Irreversibility index (0.0 fully reversible to 1.0 permanent); reparation feasibility score; intergenerational transfer coefficient

H.1.2 — Composite Score Computation

The EHI composite score is computed as a weighted product function of the five dimensions, normalized to a 0–100 scale. The Reversibility dimension receives an amplifying weight because irreversible harm is categorically distinct from remediable harm in its legal consequences under Section 5.2(a). The weighting coefficients are subject to mandatory peer review and recalibration by the OHM on a biennial basis, with all calibration data published in the annual Harm Report.

Component	Symbol	Default Weight	Rationale
Breadth	B	0.20	Scale of harm is a primary determinant of systemic vs. individual exploitation
Magnitude	M	0.25	Per-unit severity determines the human cost independent of scale
Duration	D	0.15	Persistent harm compounds; onset-to-detection lag is a governance failure signal
Concentration	C	0.15	Targeted harm against Vulnerable Parties is qualitatively worse than diffuse harm
Reversibility	R	0.25 (+amplifier)	Irreversible harm triggers the criminal liability threshold; the amplifier activates above $R > 0.7$

H.1.3 — Uncertainty Quantification

Every EHI score is issued with a documented uncertainty envelope, conforming to the methodology specified in the Guide to the Expression of Uncertainty in Measurement (GUM). This is not an optional quality measure — it is a due process requirement. An Actor cannot be held liable based on an EHI score whose uncertainty range spans a liability threshold. The OHM must report:

- (a) The Type A uncertainty components — those estimated from statistical analysis of repeated observations, including variability in harm survey data, health outcome registries, and financial loss documentation;
- (b) The Type B uncertainty components — those estimated from non-statistical means, including expert judgment, prior literature, and model parameter uncertainty;
- (c) The combined standard uncertainty (u_c) and the expanded uncertainty (U) at a 95% confidence level; and
- (d) The sensitivity coefficients showing which input variables most affect the composite score, for purposes of adversarial audit and targeted data collection.

DUE PROCESS ANCHOR — Why GUM Compliance Is Non-Negotiable

The EHI is an instrument of legal accountability. When it produces a score that triggers incarceration, the Actor subject to that score has a constitutional right to challenge the measurement's validity. GUM-compliant uncertainty reporting provides the evidentiary foundation for that challenge — and for the court's assessment of whether the measurement meets the clear-and-convincing or beyond-reasonable-doubt standard applicable to the specific liability tier. An EHI score without a documented uncertainty envelope is not admissible as the primary basis for criminal liability under Section 5.2.

H.1.4 — Statistical Process Control Thresholds

The OHM's ongoing monitoring function operates through Statistical Process Control (SPC) principles: establishing baseline distributions of sector-specific indicators, computing control limits from historical variance, and treating observations outside those control limits as signals requiring investigation. SPC-based detection serves three functions in the EHI framework:

- (a) **BASELINE ESTABLISHMENT.** For each monitored sector, the OHM maintains a rolling baseline of key harm indicators (health outcomes, financial distress rates, environmental markers, civic participation metrics). Baselines are computed from pre-SEPA historical data and updated annually.
- (b) **ANOMALY DETECTION.** Observations exceeding three standard deviations from baseline (3-sigma control limit) trigger an automated investigation flag. Observations exceeding two standard deviations (2-sigma warning limit) trigger enhanced monitoring. This is the primary mechanism by which the AI monitoring system identifies candidate exploitation events for EHI assessment.
- (c) **RECURRENCE DETECTION.** The SPC framework tracks whether a flagged Actor's harm indicators return to baseline following a remediation event, or whether they persist or accelerate. Failure to return to baseline within the remediation period specified in the enforcement order is the primary evidentiary basis for the 'recurrence without caring' finding under Section 5.2(b).

H.2 — THE FOUR-LAYER AI MONITORING ARCHITECTURE

The OHM's AI monitoring system is organized as a four-layer holonic architecture, in which each layer is simultaneously a whole system with its own internal governance and a component of the larger enforcement system above it. This architecture is chosen specifically because holonic systems are intrinsically more resistant to single-point capture: compromising any one layer does not compromise the layers above or below it without additional, independently audited breaches.

The four layers correspond to the four fundamental activities of any measurement-based governance system: signal detection (sensing the world), pattern recognition (making sense of signals), harm assessment (computing legal consequences), and enforcement communication (transmitting findings to action). Each layer has distinct data requirements, AI capabilities, human oversight requirements, and capture-resistance mechanisms.

Layer	Name	Function	Primary AI Type	Human Oversight
L1	Signal Detection	Continuous ingestion and normalization of structured and unstructured data streams from all monitored domains	Data pipeline AI; anomaly pre-filter; NLP for unstructured sources	Automated; human audit of ingestion failures and source additions
L2	Pattern Recognition	Identification of behavioral anomalies, SPC threshold crossings, and Actor-harm causal chain construction	Supervised and unsupervised ML; causal inference models; graph neural	Human-in-the-loop for any pattern that crosses the investigation threshold

Layer	Name	Function	Primary AI Type	Human Oversight
			networks for Actor network mapping	
L3	Harm Assessment	EHI computation, uncertainty quantification, and liability tier determination for flagged patterns	Structured AI reasoning over EHI model; GUM-compliant uncertainty propagation; SPC baseline comparison	Mandatory human review before any EHI score is finalized and transmitted to OEP
L4	Enforcement Communication	Transmission of finalized EHI assessments to OEP, public Harm Registry, Actor notification system, and victim notification system	Document generation; secure transmission; audit trail maintenance; public dashboard update	OHM Director sign-off required for criminal referral; automated for civil filing and public registry update

H.2.1 — Layer 1: Signal Detection and Data Ingestion

The L1 layer is the sensory infrastructure of the monitoring system. It continuously ingests data from a defined universe of structured and unstructured sources across all monitored domains, normalizes them into the common data schema required for L2 pattern recognition, and applies a pre-filter to flag anomalous observations for L2 processing. L1 does not make legal determinations — it makes no EHI assessments and generates no enforcement outputs. Its sole function is accurate, complete, and tamper-resistant data capture.

L1 data sources are classified into four tiers by reliability and verification requirements:

- (a) **TIER 1 — Official Records.** Government-maintained data with an established chain of custody: health outcome registries, financial regulatory filings, environmental monitoring stations, court records, legislative voting records, campaign finance disclosures, property transaction records, tax filings, and employment records.
- (b) **TIER 2 — Institutional Self-Reports.** Data submitted by Actors under mandatory OHM reporting requirements: corporate harm impact reports, platform engagement and outcome data, educational outcome disclosures, and healthcare quality metrics. Tier 2 data is subject to independent audit and cross-validation against Tier 1 sources.
- (c) **TIER 3 — Third-Party Research.** Peer-reviewed academic and institutional research, public health surveillance data, environmental monitoring by accredited third parties, and independent audit reports. Tier 3 data is weighted by source credibility score maintained by the OHM.
- (d) **TIER 4 — Public and Participatory Data.** Citizen complaint submissions via the OHM public portal, journalistic investigations, civil society reporting, social media pattern data (with privacy protections as specified in H.4). Tier 4 data triggers investigation but cannot independently produce EHI scores without cross-validation against Tier 1–3 sources.

H.2.2 — Layer 2: Pattern Recognition and Causal Chain Construction

L2 is the analytical core of the monitoring system. It receives pre-filtered signals from L1 and applies three distinct AI capabilities to construct a complete picture of potential exploitation events: behavioral anomaly detection, causal chain construction, and Actor network mapping.

BEHAVIORAL ANOMALY DETECTION. Unsupervised machine learning models trained on baseline distributions for each sector identify SPC threshold crossings and novel pattern types not present in training data. A flagged anomaly is not a legal finding — it is an investigation trigger. The system is calibrated to err on the side of investigation at the 2-sigma warning level, reserving the 3-sigma action threshold for automated escalation.

CAUSAL CHAIN CONSTRUCTION. Causal inference models trace the path from a detected anomaly back to Actor conduct. This is the most technically demanding function in the monitoring system: it must establish, with documented confidence intervals, that a specific Actor's conduct is a necessary and sufficient cause of the observed harm signal, rather than merely correlated with it. The causal chain output is the primary evidentiary input to L3 EHI computation.

ACTOR NETWORK MAPPING. Graph neural networks map the organizational, financial, and relational connections between Actors to identify cases where liability is distributed across an Actor network rather than concentrated in a single entity. This is essential for detecting coordinated exploitation patterns (as specified in Addendum G.5), regulatory capture chains (Addendum B), and institutional co-exploitation architectures (Addendum D).

H.2.3 — Layer 3: Harm Assessment and EHI Computation

L3 receives the complete causal chain package from L2 and computes the formal EHI score for submission to the human review panel. The L3 AI system is a structured reasoning system, not a generative model — it applies the EHI scoring framework specified in H.1 to the causal chain evidence, propagates uncertainty from each input through the composite score computation, and produces a GUM-compliant measurement report that includes the score, the uncertainty envelope, the sensitivity analysis, and the evidentiary basis for each dimension score.

The L3 system operates under a mandatory circuit-breaker architecture that halts automated processing and requires human intervention under the following conditions:

- (a) When the EHI uncertainty range spans a liability tier boundary — ensuring that boundary cases are resolved by human judgment rather than automated threshold application;
- (b) When the causal chain confidence falls below the minimum evidentiary threshold established by the OHM's legal team for the applicable liability tier;
- (c) When the Actor's legal team has filed a challenge to the investigation, triggering the L3 assessment; or
- (d) When the scoring involves a novel harm type not present in the validated EHI model, requiring model extension before scoring can proceed.

AI CIRCUIT BREAKER — When the System Stops Itself

The circuit breaker conditions above are not failure modes — they are success conditions. A measurement system that knows the boundaries of its own reliable operation and halts before exceeding them is more trustworthy than one that produces confident outputs regardless of evidential quality. The circuit breaker is the operational expression of epistemic humility in the legal accountability context: when the machine is uncertain, the human decides. This principle governs the entire L3 architecture and is non-negotiable in any SEPA-compliant implementation.

H.2.4 — Layer 4: Enforcement Communication

L4 manages the secure, auditable transmission of finalized EHI assessments to all required recipients and the maintenance of the public-facing Harm Registry. The L4 system is deliberately simpler than L1–L3: its primary technical requirements are security, auditability, and non-repudiation. Every enforcement communication is cryptographically signed, timestamped, and recorded in an immutable audit log accessible to all parties.

L4 operates four distinct communication channels, each with specific content requirements and authorization levels:

- (a) OEP REFERRAL CHANNEL. Transmits complete EHI assessment packages to the Office of Exploitation Prosecution for civil or criminal action. Requires OHM Director authorization for criminal referrals.
- (b) PUBLIC HARM REGISTRY. Publishes the EHI score, affected domain, Actor category, and remediation status for every finalized civil finding. Actor identity is published upon finalization of the enforcement action, not upon initiation of the investigation.
- (c) ACTOR NOTIFICATION SYSTEM. Provides the investigated Actor with the complete EHI assessment, uncertainty documentation, evidentiary basis, and appeal pathway within 72 hours of L3 finalization.
- (d) VICTIM AND COMMUNITY NOTIFICATION. Notifies identified Vulnerable Parties and affected communities of an enforcement finding, available remediation pathways, and their rights under the Act — in plain language, in the primary language of the affected community.

H.3 — REQUIREMENTS ENGINEERING AND PRE-DEPLOYMENT CERTIFICATION

S.E.P.A. Addendum A (Section A.2) and Addendum E (Section E.3) establish that platforms and engineers deploying autonomous systems face strict liability for the harm those systems produce, and that platforms must pass a high-assurance trust measurement certification before deployment. This section specifies the requirements engineering standard that governs both the OHM's own AI systems and the third-party systems that those AI systems are charged with monitoring.

H.3.1 — STPA/STAMP-Based Hazard Analysis for SEPA-Regulated Systems

Any AI system deployed by the OHM or subject to OHM pre-deployment certification under Section A.2 must undergo Systems-Theoretic Process Analysis (STPA) and Systems-Theoretic Accident Model and Processes (STAMP) hazard analysis as a precondition for deployment approval. This requirement is drawn from the aerospace-grade safety engineering standard and applied to the exploitation governance context.

The STPA/STAMP analysis for a SEPA-regulated AI system must identify and document:

- (a) UNSAFE CONTROL ACTIONS (UCAs). For each control action the system can take — including outputs, recommendations, flags, and non-actions — the analysis identifies the conditions under which that action is unsafe, too-early, too-late, too-long, or too-short from the perspective of the SEPA enforcement mission;

- (b) UCA CONSTRAINTS. For each identified UCA, a corresponding constraint is formally specified and implemented in the system architecture, with verification evidence that the constraint is effective.
- (c) LOSS SCENARIOS. Causal factor chains connecting controller failures, process model errors, and disturbances to each identified loss scenario — including misidentification of non-exploitative conduct as exploitation (false positive) and failure to detect actual exploitation (false negative); and
- (d) SAFEGUARDS AND MITIGATIONS. Engineered safeguards for each loss scenario, with residual risk documentation and acceptance rationale.

UCA RECORD REQUIREMENT

The OHM is required to maintain a formal UCA record for each AI system component, specifying the unsafe control action, the context in which it is unsafe, the constraint implemented to prevent it, and the verification evidence for that constraint. The UCARecord is a public document subject to an annual civil liberties audit. This requirement is the primary technical mechanism for preventing the monitoring system from becoming a tool of suppression of legitimate collective action, political organizing, or protected expression.

H.3.2 — Mean Time Between Hallucinations (MTBH) as an AI Reliability Metric

Generative and large-scale language AI systems are subject to a characteristic failure mode — producing confident, syntactically coherent outputs that are factually incorrect, internally inconsistent, or unsupported by the evidence provided. In the legal accountability context, this failure mode has a specific name: hallucination. A system that hallucinates EHI scores, fabricates causal chain evidence, or misattributes harm to an Actor is not merely unreliable — it is a direct threat to the due process rights of the Actors it monitors and the integrity of the enforcement system.

The OHM is required to specify, measure, and publicly report Mean Time Between Hallucinations (MTBH) as a primary reliability metric for any AI system component that produces legal outputs. MTBH is defined as the mean interval between independently verified instances of a system producing an output that is factually unsupported by its input evidence. Requirements for MTBH implementation:

- (a) MTBH is measured through continuous red-team testing by an independent evaluation team with no organizational connection to the OHM's technology operations;
- (b) AI system components with MTBH below the minimum threshold established for their liability tier are immediately suspended from production use pending remediation;
- (c) MTBH values are published in the Annual Harm Report and subject to independent verification; and
- (d) Vendors who supply AI components to the OHM are contractually liable for MTBH performance below specified thresholds, with liquidated damages calibrated to the liability tier of the system component.

H.3.3 — Third-Party Platform Certification

Any platform deploying an AI system at scale whose operational design has a plausible pathway to S.E.P.A. liability — as specified in Addenda A.2 (Technology Platforms), E.3 (Engineers), and F.4 (EdTech) — must obtain OHM certification before deployment and renew it annually. Certification requires:

- (a) Submission of the complete STPA/STAMP hazard analysis, UCAR record, and MTBH measurement methodology to the OHM;
- (b) Independent third-party audit of the system's EHI-relevant behavioral outputs using OHM-specified test protocols;
- (c) Real-time behavioral telemetry sharing with the OHM's L1 monitoring layer, subject to privacy protections specified in H.4; and
- (d) Continuous EHI monitoring by the OHM's L2 layer, with the platform's certification suspended automatically upon L3 confirmation of an EHI score above the civil liability threshold.

H.4 — AI SYSTEM SAFEGUARDS AND CAPTURE RESISTANCE

The monitoring architecture described in H.2 represents a concentration of observational power that, if captured or misused, could itself constitute a form of exploitation at the civilizational scale. A surveillance system designed to detect exploitation but deployed to suppress dissent, monitor political organizing, or extract behavioral data for state or commercial benefit would be the most efficient exploitation mechanism ever constructed. The safeguards in this section are not optional features — they are the constitutional floor of the monitoring architecture.

H.4.1 — Scope Limitation and Purpose Binding

The OHM's AI monitoring systems are purpose-bound by statute to the detection and measurement of exploitation as defined in S.E.P.A. They may not be used, queried, or adapted for any other purpose without an Act of Congress specifying the additional purpose, subject to the same two-thirds/three-quarters amendment requirement established for the ERTF in Section 6 of the core statute. Specifically prohibited secondary uses include:

- (a) Immigration enforcement or population tracking;
- (b) Political surveillance, monitoring of protected organizing activity, or tracking of association between persons absent a specific, documented exploitation investigation;
- (c) Commercial data analysis or any data sharing with private entities not involved in a specific, active OHM investigation; and
- (d) Intelligence operations by any executive branch agency.

H.4.2 — Adversarial Robustness Requirements

Actors subject to monitoring under S.E.P.A. have strong financial incentives to game, evade, or corrupt the monitoring system. The OHM's AI architecture must be designed with adversarial actors as the baseline threat model, not as an edge case. Required adversarial robustness measures include:

- (a) **EVASION RESISTANCE.** The L2 pattern recognition system must be tested against adversarial evasion attacks — structured attempts to modify Actor behavior to fall just below detection thresholds while preserving the exploitative outcome — and must maintain detection performance above the minimum threshold specified for each domain.
- (b) **DATA POISONING RESISTANCE.** Tier 2 self-report data is cross-validated against independent Tier 1 and Tier 3 sources before use in EHI computation. Systematic discrepancies between self-reported and independently verified data are themselves treated as investigation triggers.
- (c) **MODEL MANIPULATION RESISTANCE.** The OHM's AI model weights, training data, and calibration parameters are classified as critical infrastructure and protected against unauthorized modification by the same legal framework that protects federal election infrastructure; and
- (d) **INSTITUTIONAL CAPTURE RESISTANCE.** The technology operations team responsible for the OHM's AI systems is subject to the same revolving-door prohibitions and appointment insulation requirements as OHM officers generally, as specified in Section 4.2 of the core statute.

H.4.3 — Privacy Architecture

The monitoring system's data collection operates under a data minimization principle: the OHM collects only the data necessary to compute EHI scores for specific investigation targets, and retains it only for the duration of the investigation plus the statutory appeals period. The system architecture enforces the following privacy protections by design:

- (a) Individual-level data is aggregated before L2 pattern analysis wherever aggregate-level detection is sufficient for the relevant harm type;
- (b) Individual identities are not exposed at the L1 or L2 layers; de-anonymization occurs only at L3 upon confirmation of a specific investigation target, subject to a documented legal authorization requirement.
- (c) All L1 data streams from private communications, medical records, or financial transactions require a judicially issued warrant based on evidence of a specific investigation target and a documented harm signal exceeding the 2-sigma warning threshold; and
- (d) An annual privacy audit by an independent panel including representatives of civil liberties organizations is a mandatory publication requirement, with results included in the Annual Harm Report.

H.4.4 — The Morphism Integrity Requirement

The OHM's monitoring system must maintain structural consistency between its representation of exploitation patterns and the actual conduct it purports to measure. In systems engineering terms, this is a morphism requirement: the mapping from observed Actor behavior to computed EHI score must preserve the structure of the underlying harm relationship. A score that results from a monitoring artifact — a data pipeline error, a model bias, a measurement instrument failure — is not a valid measurement of exploitation harm; it is a measurement of the monitoring system's own error.

The morphism integrity requirement is operationalized through three mechanisms:

- (a) **CONSTRUCT VALIDITY TESTING.** Each EHI dimension score is validated against at least two independent measurement approaches before being incorporated into a composite score. Where the approaches diverge by more than the expanded uncertainty threshold, the dimension score is flagged for human review.

(b) PREDICTIVE VALIDITY MONITORING. The OHM tracks whether EHI scores predict subsequent harm outcomes as expected — whether high-scoring enforcement actions produce measurable harm reduction upon remediation. Systematic failure of predictive validity triggers mandatory model recalibration.

(c) CROSS-DOMAIN CONSISTENCY AUDITS. The EHI scoring methodology must produce consistent results for structurally equivalent harm patterns across different domains. A scoring system that rates identical extraction-from-vulnerability patterns differently depending on whether the Actor is a corporation (Addendum A) or a religious institution (Addendum D) fails the morphism integrity requirement and must be recalibrated before continued use.

H.5 — ENFORCEMENT COMMUNICATION ARCHITECTURE

The value of a measurement system is realized only when its outputs produce action. The enforcement communication architecture specified in this section defines the complete pathway from L4 transmission to enforcement action, public accountability, and victim notification — including the due process guarantees that govern the Actor's right to challenge OHM findings before enforcement consequences attach.

H.5.1 — The Violation Communication Protocol

Stage	Trigger	Responsible Party	Output and Timeline
1 — Detection	L1/L2 SPC threshold crossing	Automated (L2 AI)	Investigation flag; internal notification to the OHM investigative team. 24 hours.
2 — Investigation	Investigation flag acceptance by the OHM team	OHM investigative team + L2 AI	Causal chain construction; Tier 1–3 data collection; initial EHI pre-score. 30–90 days.
3 — Assessment	Completed causal chain package	L3 AI + mandatory human review panel	GUM-compliant EHI score; uncertainty envelope; liability tier determination. 15–30 days.
4 — Actor Notice	L3 finalization	L4 Actor Notification System	Full assessment package + appeal rights delivered to Actor. 72 hours.
5 — Appeal Window	Actor notices delivery	OHM Appeals Panel (independent)	Actor may submit evidence, challenge methodology, or request re-measurement. 60 days.
6 — Finalization	Appeal resolution or window expiry	OHM Director	Final EHI score with appeal disposition. Director authorization for criminal referral.
7 — Enforcement	Finalization	OEP (civil/criminal) + L4 Registry	OEP filing; Public Harm Registry publication; victim/community notification. 30 days.

H.5.2 — The Public Harm Registry

The Public Harm Registry is the primary transparency mechanism of the enforcement architecture. It is a publicly accessible, machine-readable, immutable ledger of all finalized enforcement findings under S.E.P.A. The Registry is designed to serve multiple functions simultaneously:

- (a) **ACCOUNTABILITY TRANSPARENCY.** Citizens, journalists, researchers, and civil society organizations can query the Registry by Actor, sector, geographic area, EHI score range, harm type, and remediation status.
- (b) **DETERRENCE SIGNAL.** The public visibility of EHI scores and enforcement actions functions as a reputational consequence independent of civil or criminal penalties, altering the decision calculus of Actors who are sensitive to reputational harm;
- (c) **RESEARCH INFRASTRUCTURE.** The Registry's anonymized aggregate data is a mandatory open data resource for academic research on exploitation patterns, remediation effectiveness, and EHI model performance; and
- (d) **RECURRENCE TRACKING.** The Registry is the authoritative source for 'prior adjudication' determinations under Section 5.2(b) of the core statute. Any Actor with a Registry entry is on constructive notice that recurrence triggers criminal liability.

H.5.3 — AI-Generated Plain-Language Communication

The technical outputs of the EHI process — GUM-compliant measurement reports, causal chain documentation, Actor notification packages — are necessarily complex. The L4 system is required to produce, alongside every technical output, a plain-language summary designed for non-specialist readers: affected community members, journalists, and the general public. These summaries must:

- (a) Describe the exploitative act in plain language, without technical jargon, in a manner consistent with the legal findings;
- (b) Explain the harm in human terms — how many people were affected, in what ways, for how long, and whether it is reversible;
- (c) Identify the remediation required and the timeline for compliance; and
- (d) Be translated into the primary languages of the affected communities, with translation quality verified by community review.

AI-generated plain-language summaries are reviewed by the OHM's communications team before publication. They cannot deviate from the technical findings in any legally material way — they are summaries, not independent assessments.

H.6 — CROSS-DOMAIN MONITORING PROFILES: ADDENDA A–G

This section provides monitoring profile specifications for each prior Addendum domain, translating the behavioral patterns and liability standards in Addenda A through G into the concrete data sources, detection methods, EHI dimension engagement, and enforcement outputs that the AI monitoring architecture will produce in each domain. These profiles are the operational interface between the legal framework and its technical implementation.

Addendum A — Economic Sectors and Markets

MONITORING PROFILE: ADDENDUM A: ECONOMIC SECTORS

Data Sources: SEC filings; corporate financial statements; Bureau of Labor Statistics workforce data; Federal Reserve financial stability reports; EPA environmental monitoring stations; OSHA incident records; CDC community health outcome registries; bankruptcy filing data; pension fund disclosures; credit bureau aggregate distress metrics.

Detection Method: Graph neural network Actor-network mapping to trace asset movement across corporate structures; time-series anomaly detection on community health outcomes correlated with corporate transactions; causal inference models linking PE acquisition events to subsequent health outcome degradation; SPC monitoring of wage-to-productivity ratios for wage suppression signals.

EHI Dimensions Engaged: Breadth (affected workforce/community population size); Magnitude (per-capita financial loss, health outcome decline); Duration (onset of extraction to detection); Concentration (geographic and demographic targeting of harm); Reversibility (pension destruction, permanent job loss, ecosystem damage — high R scores).

Enforcement Output: Civil EHI reports to OEP for financial harm patterns; criminal referral packages for irreversible community health harm linked to specific executive decisions; Public Harm Registry entries on all finalized corporate exploitation findings; community remediation fund activation triggers.

Addendum B — Government: Extractive Exploitation

MONITORING PROFILE: ADDENDUM B: GOVERNMENT (EXTRACTIVE)

Data Sources: Congressional voting records; FEC campaign finance disclosures; lobbying disclosure filings (HLOGA); regulatory agency decision logs; executive appointment records; agency budget allocation data; government contract award databases; legislative text with natural language processing for corporate-authored provisions; judicial financial disclosure reports.

Detection Method: NLP analysis of legislation text to identify provisions with narrow beneficiary profiles inconsistent with stated public purpose; correlation modeling between lobbying expenditure patterns and legislative outcomes; causal chain construction from regulatory capture events (appointment records, agency budget cuts) to measurable public harm; financial disclosure anomaly detection for judicial gift patterns.

EHI Dimensions Engaged: Breadth (affected public population); Magnitude (health/financial/civic harm per capita from policy); Duration (legislative harm persists until repeal — extended D scores); Concentration (targeted harm to specific demographics or communities); Reversibility (irreversible harm from regulatory rollbacks enabling industrial incidents).

Enforcement Output: OEP referral packages for legislators whose voting records show statistically significant correlation between lobbying receipts and documented population harm; judicial misconduct referrals for gift-pattern anomalies; Public Harm Registry entries for regulatory capture events with measurable public harm.

Addendum C — Government: Non-Extractive Exploitation

MONITORING PROFILE: ADDENDUM C: GOVERNMENT (NON-EXTRACTIVE)

Data Sources: Redistricting records and mathematical analysis; hate crime and violence incident databases correlated with political messaging; emergency response timing data; CDC public health statement archives vs. internal communications (subpoena-accessible); FOIA-released agency

documents; classified data declassification pipeline (per court order); docket management records with outcome analysis.

Detection Method: Geospatial analysis of redistricting maps against population and demographic data to detect mathematical disenfranchisement signatures; sentiment and content analysis of political communications correlated with subsequent hate crime data; time-series analysis of emergency response data against known crisis onset dates; natural language processing of public health statements for material inconsistencies with contemporaneous internal records.

EHI Dimensions Engaged: Breadth (disenfranchised voter population; communities exposed to false safety assurances); Magnitude (democratic representation harm; physical harm from suppressed health data); Duration (gerrymander persists across election cycles; information suppression harm accumulates); Concentration (targeted at specific political or demographic groups); Reversibility (foregone elections are irreversible; preventable illness deaths are irreversible).

Enforcement Output: OEP referral for redistricting maps meeting mathematical disenfranchisement threshold; criminal referral for executive branch officials where subpoenaed internal records demonstrate deliberate public safety information suppression; judicial misconduct packages for docket management anomalies.

Addendum D — Organizations by Type

MONITORING PROFILE: ADDENDUM D: ORGANIZATIONS

Data Sources: IRS Form 990 filings (nonprofit financial transparency); corporate product safety databases; FDA adverse event reporting; FTC consumer complaint data; CPSC product recall records; social media engagement and mental health outcome data (aggregated, privacy-protected); religious organization financial data (where tax-exempt status creates reporting obligations); state attorney general charitable registration data.

Detection Method: For-profit: product harm signal detection using FDA adverse event and FTC complaint clustering; algorithmic outcome analysis for platforms claiming engagement optimization. Nonprofit: outcome-to-expenditure ratio monitoring against stated mission metrics; lobbying activity correlation with organizational mission. Religious: financial extraction pattern detection against congregant income data (aggregated); institutional response-time analysis for abuse report patterns.

EHI Dimensions Engaged: Breadth (consumer population harmed; beneficiary population served vs. exploited; congregant financial harm); Magnitude (psychological harm per user from platform design; financial extraction from vulnerable congregants); Duration (persistent product harm; sustained mission non-delivery); Concentration (vulnerable population targeting by religious and nonprofit actors); Reversibility (psychological harm from addictive platform design — moderate-to-high R).

Enforcement Output: Civil EHI reports for for-profit product harm; asset seizure and dissolution referrals for nonprofit mission-exploitation patterns; IRS tax-exempt status revocation referrals for religious organizations with documented financial exploitation patterns; OEP criminal referrals for institutional abuse cover-up patterns.

Addendum E — Licensed and Expert Professions

MONITORING PROFILE: ADDENDUM E: PROFESSIONS

Data Sources: Medical licensing board records; DEA prescribing data; CMS claims data (Medicare/Medicaid); malpractice claim registries; pharmaceutical payment tracking databases (Open Payments); court docket and case outcome records; prosecutorial conviction rate and misconduct tracking; engineering project failure databases; scientific retraction registries; journalistic correction and misinformation tracking databases.

Detection Method: Prescribing pattern anomaly detection using SPC baselines for physician specialty and patient population (detecting opioid over-prescribing signatures); civil litigation outcome pattern analysis for legal exploitation signatures (plaintiff resource exhaustion vs. case merit metrics); engineering certification seal correlation with subsequent failure events; scientific publication network analysis for coordinated manufactured consensus patterns; media content analysis for documented disinformation deployment correlated with public health outcome data.

EHI Dimensions Engaged: Breadth (patient population harmed; communities affected by wrongful convictions; public misled by manufactured science); Magnitude (addiction severity; years imprisoned; climate harm scale for fossil fuel disinformation); Duration (prescribing pattern duration; length of wrongful imprisonment); Concentration (vulnerable patient populations; communities subject to prosecutorial misconduct); Reversibility (addiction harm — high R; wrongful imprisonment — irreversible; climate trajectory lock-in — high R).

Enforcement Output: Medical board license revocation referrals; OEP criminal referral packages for prosecutors with documented exculpatory evidence suppression; engineering seal suspension for certified failures; research funding revocation referrals for manufactured consensus patterns; OEP civil filings for media exploitation of provable disinformation causing documented public harm.

Addendum F — Education

MONITORING PROFILE: ADDENDUM F: EDUCATION

Data Sources: National Center for Education Statistics (NCES) district-level funding and outcome data; juvenile justice intake data cross-referenced with school disciplinary records; standardized test result distributions by socioeconomic demographic; university endowment and tuition data against graduate wage outcome databases; student loan default rates by institution and program; EdTech data broker transaction records (via mandatory OHM reporting); Title IX complaint and resolution records.

Detection Method: SPC-based monitoring of educational outcome inequality across funding distribution — detecting zip-code-correlated performance gaps exceeding control limits; school-to-prison pipeline signal detection using disciplinary record to juvenile intake correlation; university ROI analysis using program-level graduate wage data against tuition extracted; EdTech data broker network analysis for minor student data transaction patterns; Title IX resolution timeline analysis for institutional suppression signatures.

EHI Dimensions Engaged: Breadth (student population affected by structural funding inequality; graduates carrying negative-ROI debt); Magnitude (lifetime earnings loss from educational deprivation; debt burden per student); Duration (funding inequality persists across decades — high D score); Concentration (geographic and demographic targeting of educational harm — high C for property-tax funding systems); Reversibility (intergenerational harm from educational deprivation — very high R; EdTech data harm to minors — high R).

Enforcement Output: State legislature referrals for funding structures meeting mathematical inequality threshold; administrator incarceration referrals for documented school-to-prison pipeline patterns; university endowment reparation orders for negative-ROI credential programs; EdTech executive criminal referrals for minor data harvesting; Title IX investigation referrals for institutional suppression patterns.

Addendum G — Individual Citizens

MONITORING PROFILE: ADDENDUM G: INDIVIDUAL CITIZENS

Data Sources: Platform transparency reports on coordinated inauthentic behavior; emergency service dispatch records with demographic cross-referencing; local zoning board meeting records and

outcome data with housing availability correlation; MLM financial structure disclosures (FTC); civil court records for domestic financial exploitation patterns; public health platform content analysis (aggregated, privacy-protected); 911 call pattern analysis (demographic correlation, jurisdictional aggregation).

Detection Method: Network topology analysis of coordinated harassment campaigns (timing, content similarity, amplification path) without individual surveillance; 911 call demographic correlation analysis using jurisdictional aggregation — detecting racial targeting signatures at population level without individual monitoring; housing availability time-series correlation with NIMBY coalition procedural activity records; MLM recruitment network analysis for pyramid structure identification; public health content analysis for deliberate disinformation deployment correlated with vaccine uptake or treatment-seeking data.

EHI Dimensions Engaged: Breadth (campaign victim count; housing-insecure population affected by NIMBY blocking; MLM victim network size); Magnitude (psychological harm severity; financial loss per MLM victim; housing harm per affected household); Duration (campaign duration; years of housing blocking; MLM recruitment period); Concentration (targeted demographic harassment; specific communities blocked from housing); Reversibility (psychological trauma from coordinated harassment — moderate-to-high R; foregone housing — moderate R).

Enforcement Output: Platform co-liability referrals for coordinated harassment amplification; civil liability packages for documented 911-call racial targeting patterns; NIMBY coalition financial liability assessments for housing harm; MLM structure civil enforcement referrals; aggregate harm criminal referrals for campaign organizers meeting EHI criminal threshold; de minimis determinations protecting below-threshold individual conduct.

IMPLEMENTATION NOTE — The Aggregate Harm Monitoring Bridge

The Addendum G monitoring profile is the most technically constrained in the framework because individual digital and civic conduct is the domain where the risk of over-surveillance is highest and the protection of legitimate collective action is most critical. The monitoring approach specified here operates exclusively at the aggregate and network topology level — it detects coordinated patterns, not individual behavior. Individual de-anonymization occurs only upon L3 confirmation of specific Actor involvement in a pattern that meets the EHI civil threshold, subject to the warrant requirement specified in H.4.3. The OHM's annual civil liberties audit must specifically evaluate whether the G-domain monitoring is producing false positives against legitimate organizing activity.

ADDENDUM H — CONCLUSION: THE MEASUREMENT IMPERATIVE

The Systemic Exploitation Prevention Act is, at its technical foundation, a measurement law. Its central claim — that exploitation is a behavioral anomaly, not a matter of intent, and that the behavioral anomaly is identifiable, quantifiable, and attributable — is only legally and operationally meaningful if the measurement infrastructure is equal to that claim. A prohibition without a measurement system is a statement of aspiration. A measurement system without capture resistance is a tool of oppression. A measurement system without due process protections is an instrument of persecution.

This Addendum has specified the architecture required to make S.E.P.A.'s central claim real: a five-dimensional EHI grounded in GUM-compliant uncertainty quantification and SPC-based anomaly detection; a four-layer holonic AI monitoring architecture whose operational logic is transparency, accuracy, and structural resistance to capture; requirements engineering standards

borrowed from aerospace-grade safety practice; privacy protections and civil liberties constraints that prevent the monitoring system from becoming the harm it was built to detect; and cross-domain monitoring profiles that translate the law's prohibitions into the specific data streams, detection methods, and enforcement outputs that will determine whether those prohibitions have any effect on the world.

Three final observations are necessary:

OBSERVATION 1 — THE EHI IS NEVER FINISHED. The Exploitation Harm Index is not a document that is written once and consulted forever. It is a living measurement instrument that must be continuously calibrated against new harm patterns, new Actor strategies for evading detection, new scientific understanding of harm mechanisms, and new social contexts that the original drafters did not anticipate. The OHM's mandatory biennial recalibration requirement is not administrative overhead — it is the primary mechanism through which the law remains capable of governing the future.

OBSERVATION 2 — AI IS A TOOL, NOT A JUDGE. Nothing in this Addendum authorizes or intends for AI systems to make legal determinations. The circuit breaker requirements, the mandatory human review provisions, and the OHM Director authorization requirements for criminal referrals are not merely procedural safeguards — they are an expression of a fundamental principle: the law is a human institution, accountable to human beings, and the decision to hold a person or institution legally responsible for harm must ultimately be made by a human being who can be held accountable for that decision. AI accelerates detection, improves measurement, and scales enforcement. It does not replace judgment.

OBSERVATION 3 — THE MONITORING SYSTEM IS ITSELF SUBJECT TO THE LAW. The OHM, the OEP, and the AI systems they operate are Actors under S.E.P.A. If the monitoring architecture produces false EHI scores that damage innocent parties, if it is deployed to suppress protected expression or organizing activity, if its operators exploit the informational asymmetry between the monitored and the monitor for institutional benefit — these are acts of exploitation subject to the same strict liability standard as any other Actor. The law that the monitoring system enforces applies to the monitoring system itself. This is not an irony or a paradox. It is the necessary condition for any governance architecture that takes its own principles seriously.

Section	Component	Key Technical Standard	Primary Safeguard
H.1	Exploitation Harm Index	Five-dimensional composite; GUM uncertainty; SPC thresholds	Due process anchor — uncertainty envelope required for all legal outputs
H.2	Four-Layer Architecture	Holonic design; L1–L4 separation of functions	Circuit breaker — system halts at uncertainty/threshold boundaries
H.3	Requirements Engineering	STPA/STAMP hazard analysis; UCAREcord; MTBH metric	Pre-deployment certification mandatory for SEPA-regulated AI systems
H.4	AI Safeguards	Purpose binding; adversarial robustness; morphism integrity	Annual civil liberties audit; OHM systems subject to SEPA itself
H.5	Enforcement Communication	7-stage violation protocol; Public Harm Registry; plain-language output	60-day Actor appeal window; Director authorization for criminal referral
H.6	Domain Profiles A–G	Domain-specific data sources, detection methods, and EHI dimensions	G-domain aggregate-only monitoring; de-anonymization warrant requirement